

Phase 1: AI Voice Detection Validation

1. Objective

The objective of Phase 1 is to validate the feasibility of distinguishing human speech from synthetic (AI-generated) speech within real-world UK telephony conditions.

This phase is designed to provide a clear, evidence-based answer on whether a production-grade system is viable before committing to full-scale deployment.

2. Scope of Work

Dataset

- 50,000 - 80,000 real UK call recordings
- Telephony format: G.711 μ -law, 8kHz, mono
- Caller-only audio (no agent channel, no overlap)

Synthetic Data

- Synthetic voice samples generated using tools such as ElevenLabs
 - Inclusion of multiple voice styles and speaking patterns
 - Focus on realistic, real-world attack scenarios
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3. Approach

Data Preparation

- Audio normalization and preprocessing
- Segmentation and quality filtering
- Feature extraction aligned with telephony constraints

Model Development

- Baseline models using architectures such as Wav2Vec2 and AASIST
- Fine-tuning specifically for low-bitrate telephony audio

Evaluation Framework

- Controlled validation using real vs synthetic datasets
 - Testing across varying accents, noise conditions, and audio quality
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4. Deliverables

At the end of Phase 1, the following will be delivered:

- Trained baseline detection model
 - Evaluation report including:
 - Overall accuracy
 - False positive and false negative rates
 - Performance across accents and audio conditions
 - Synthetic test dataset used for validation
 - Summary of preprocessing and dataset strategy
 - Recommendation on production feasibility and next steps
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5. Timeline

Estimated duration: **4–5 weeks** from dataset access and environment setup.

6. Exclusions (Phase 2)

The following are not included in Phase 1 and can be scoped separately:

- Real-time API development
- Production deployment and hosting
- Monitoring and scaling infrastructure
- Integration with live call systems

7. Optional Add-ons

- Expanded dataset coverage (100k+ recordings)
- Multi-engine synthetic voice testing
- Advanced adversarial testing
- Real-time prototype system

8. Key Considerations

- Model performance may vary across accents and audio quality
- Synthetic voice diversity is critical for generalization

- Short-duration audio may impact detection confidence